Don Usitha Mihiranga Uduwakaarachchi

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Education

Master of Science in Physics **August 2023 – May 2025**

Binghamton University, New York, United States

- **Grade Point Average:** 3.957/4.000

Bachelor of Science (Hons) in Physics **January 2017 – May 2021**

University of Colombo, Sri Lanka

- **Class Obtained:** Second Class (Upper Division)
- **Grade Point Average:** 3.50

Research Interests

- Biomedical Imaging
- Machine Learning and Deep Learning
- Medical Image Processing
- Medical Image Reconstruction

Research Experience

Graduate Research Work **August 2024 - May 2025**

Department of Physics, Binghamton University

- **Title:** ***High Harmonics Generations (HHG) in Su-Schrieffer-Heeger (SSH) Chain***
- This work aims to obtain High Harmonic Spectra in SSH chain in the presence of a strong-field pulse.
- The spinless fermion 1D basis was used to construct the SSH Hamiltonian. The QuSpin Python package was employed as it allows defining the Hamiltonian in terms of second quantization and is convenient for exact diagonalization.
- The initial eigenstates of the L(=100)-dimensional Hamiltonian were obtained, and the lowest L/2 energy states were evolved in time using the Crank-Nicolson approximation, which had already been implemented, along with QuSpin methods.
- Expectation values of position were calculated during the duration of the laser pulse, and high harmonic spectra were obtained.
- **Key Skills:** Computational Modeling, Python, Attosecond Physics, Data Analysis

Quantum Mechanics 2 - Final Project **January 2024 - May 2024**

Department of Physics, Binghamton University

- **Title:** ***Exploring Quantum Geometric Tensor: Understanding Rashba Hamiltonian under Time-Reversal Symmetry and its Violation***
- This study aims to utilize formalism to derive Quantum Geometric Tensors (QGT) for non-degenerate Eigen states, exploring their behavior in parameter space through adiabatic evolution, and study the Rashba Hamiltonian under Time- reversal symmetry and the broken time-reversal symmetry.
- QGT components of Rashba Hamiltonian without external magnetic field were obtained and it gives zero Berry curvature as time-reversal symmetry and the combined inversion and spin-flip symmetry leads it to be zero. Whereas, with an external magnetic field results claimed non-trivial quantum geometry.

- **Key Skills:** Quantum Mechanics, Mathematical Modeling, Computational Physics, Python, Advanced Data Analysis

Undergraduate Research Project — April 2020 - April 2021

Department of Physics, Faculty of Science, University of Colombo, Sri Lanka

- **Title:** ***Development of a Schlieren imaging module to measure dynamic quantities in transparent fluids***
- A Schlieren module was constructed using a concave mirror and Schlieren images were taken to visualize optical inhomogeneities in transparent media which form refractive index gradients.
- Schlieren technique was used to measure dynamic quantities in fluids. Hot air flows and heat convection fields were tested using a Schlieren module.
- Image processing techniques and Optical flow detecting techniques were applied to analyze Schlieren images to obtain vector fields.
- Furthermore, it was checked whether a tested flow is turbulent or laminar using velocity vector fields.
- **Key Skills:** Image Processing, Optical Imaging, Data Analysis, Experimental Physics, MATLAB

Work Experience

Graduate Teaching Assistant — August 2023 - May 2025

Binghamton University, New York, USA

- Currently working as a discussion teaching assistant for PHYS 122 General Physics 2 in Spring 2025.
- Worked as a discussion teaching assistant for PHYS 131 General Physics 1 (Calculus-based) in Summer 2024, PHYS 121 General Physics 1 in Fall 2023 and Fall 2024, and PHYS 122 General Physics 2 in Spring 2024.

Teaching Assistant — September 2022 - June 2023

Sri Lanka Technological Campus, Sri Lanka

- Worked as a teaching assistant for the ***Classical Physics*** course, ***Introduction to Quantum Mechanics*** course, and ***Electromagnetism*** course for undergraduate students.

Teaching Assistant — May 2021 - May 2022

Department of Physics, Faculty of Science, University of Colombo, Sri Lanka

- Worked as a teaching assistant for the ***Electromagnetic Theory*** course for level 2 undergraduate students.
- Worked as a teaching assistant for the ***Physics of Semiconductor Devices*** course for level 2 undergraduate students.
- Developed new physics laboratory experiments under the guidance of senior lecturers in charge of level 1, (***Physics Laboratory*** course) and level 2, (***Electronics and Computing Laboratory*** course and ***General Physics Laboratory*** course). Worked as a laboratory instructor for these courses and graded students' reports and answer sheets. Conducted Viva Voce examinations for undergraduates.
- Worked with a team of teaching assistants to develop an online physics experiment series for High School students who follow the physical science stream

Technical Skills

- **Programming Languages**: Python, C, C++
- **Mathematical Packages**: MATLAB, Octave
- **Circuit Design and Simulation**: Multisim
- **PCB Design**: Eagle
- **Other Applications**: MS Office
- **Image Processing** and **Numerical Analysis**

Awards and Achievements

Participated in the ROBOFEST 2018 in the university category
The annual robotic competition, organized by the Faculty of Engineering, Sri Lanka Institute of Information Technology.

Bronze medal at Sri Lankan Physics Olympiad Competition - 2015
National Physics Olympiad competition organized by the Institute of Physics, Sri Lanka

Certificate of High Distinction at the Australian National Chemistry Quiz - 2014
In the Senior Division at Australian National Chemistry Quiz organized by the Royal Australian Chemical Institute.

Certificate of High Distinction at the Australian National Chemistry Quiz - 2012
In the Junior Division at Australian National Chemistry Quiz organized by the Royal Australian Chemical Institute.

Languages

- **Sinhala**: Native Language
- **English**: Proficient **(IELTS score: 7.0)**